Anthony Ibarra | Electrical & Computer Engineer

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Summary

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective career. Showcasing a strong focus in Embedded Systems, Control theory, Robotics, including pattern recognition involving AI CNN machine learning. And power electronics making me a flexible Embedded Software Engineer with a strong foundation. I possess various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness to depend on one another.

EDUCATION

University of Illinois Chicago (UIC) — Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering

Relevant Coursework: Intro to Filter Synthesis, Adaptive Digital Filters, Electromechanical Energy Conversion, Electromagnetic Compatibility, Mechatronics Embedded Design

University of Illinois Chicago (UIC) — Aug 2017 - May 2022

Bachelor of Science in Computer Engineering

Relevant Coursework: Artificial Intelligence I, Principles of Modern Control & Principles of Auto Control, Pattern Recognition I, Computer Comm Networks I, Robotics: Algorithm/Control, Embedded Systems

SKILLS

Computer programming: Python, C/C++/C# , object oriented programming, Assembly, Ubuntu, AI (artificial intelligence) & ML(machine learning)

Software Knowledge: MATLAB, Solidworks, Altium, Ltspice, HFSS Ansys simulator, Quartus

Speach: B1 Proficient in spanish

Academic Projects

Series Brushed DC motor modeling | Electromechanical Energy Conversion — Aug 2024 - Dec 2024

- Implement a behavior model of a DC motor using knowledge magnetic circuits and transformers; DC machines; and three-phase AC synchronous and induction machines
- DM – 100A DC Machine 0.3 kW (0.4 hp), 125 V, 2.4 A, and 1800 RPM; motor testing and verification, working with motors of up to 120v using 0 – 125 volt variable and 0 – 150 volt DC voltmeter /ammeters
- AC 120VA 60Hz transformer testing through AC measurements using wattmeters, voltage meters, and ammeters
- LTSpice design and simulation of DC motor mechanical behavior through electrical simulation

Optimization on Three Coil Long Range (WPT), Electromagnetic Compatibility — Jan 2024 - May 2024

- Electromagnetic Compatibility testing using a spectrum analyzer, Oscilloscopes for EMC and EMI testing for PCB and signal integrity using RF SMA coaxial cables, SMA terminators
- Create a three coil configuration for long range wireless power transfer for simulation, confirming data from research papers through simulations
- Mathematically simulated plot from Matlab regarding the Three coils as shown within the research paper
- Simulation and plot from Ansys HFSS 3D EM simulation software, simulating the three coil behavior in different configurations
- Make an IEEE formatted report using Latex, in a professionally typeset manner demonstrating discoveries and results

Self Driving Car, Mechatronics Embedded Design — Jan 2023 - May 2023

- Lead the team and manage all time constraint tasks, development and design tasks, make timely decisions for the team's success in completing the product and hardware development
- Design a motor driver circuit through FET Driver implementation, either of (single fet, half bridge, or full H-bridge), controlled via a PWM input signal from the microcontroller
- Design a Boost Converter for the implementation multiple power systems
- Product testing using Oscilloscope, multimeters, benchtop powersupplies; and Soldering fixes when necessary

- Develop and tune controllers for both the steering and velocity controls (Servo motor, DC motor) respectively
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Solder all through hole and SMT components including masking for PCB and Prototype; create any needed jumper wires for connections through crimping and soldering
- Prototype implementation of a perf board circuit, also used as an emergency backup board for PCB failure ensuring continuation of product development
- Implement Sensors and Encoders a filter for the line Camera or velocity measured input through embedded software development and test through debugging microcontroller software

Automated Watering System, Senior Design — August 2021 - May 2022

- Work in a four-student team to prototype a product implementing a automated watering systems plants by taking moisture levels, outdoor weather conditions, and plant information data into account
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Embedded programming for a Arduino Nano iot 33 to consider watering decisions per plants moisture levels and water consumption
- Create mobile APP (Kivy framework) to display necessary information regarding the system, plants moisture levels and weather conditions
- Create a UDP client-server connection for communication between powered product and the mobile application; providing manual control over a wireless connection
- Implement wireless communication through Bluetooth Protocol (BLE) for consumer and their device